Accessible Documents in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.6

Richard Stanton
richard.stanton@berkeley.edu
UC Berkeley

May 6, 2026

1 Introduction

On January 8, 2026, we were notified by campus that, beginning in April 2026:

> "The updated requirements of the ADA require that digital course materials provided to students, even materials inside password-protected course sites like bCourses, will need to comply with accessibility standards (Web Content Accessibility Guidelines (WCAG) 2.1 Level AA)."

Many of us use LaTeX to create teaching materials—both slides and documents. Standard LaTeX (including Beamer) does not automatically generate accessible PDFs.

2 What is this project?

An experiment exploring accessible LaTeX documents using the LaTeX Tagging Project:

- Two templates: Articles (`article` class) and slides (`ltx-talk`)
- Shows: What's common to both, what's different
- Contains: Working examples with math, text, graphics, and tables
- Scores: Perfect 100% from the bCourses accessibility checker (Ally)

2.1 How to use it

1. **Learn the common requirements:** Apply to any LaTeX document
2. **See class-specific needs:** Understand slides vs. articles
3. **Copy and adapt:** Use as templates for your documents
4. **Study the code:** Heavily commented for learning

Available at: https://github.com/rhstanton/accessible_LaTeX

2.2 Converting to accessible LaTeX: Overview

- **Your existing LaTeX skills transfer** - you're just adding accessibility features
- **The changes are minimal** and follow consistent patterns

Common requirements (both slides & articles):

- Add document metadata and enable PDF tagging
- Tag images with alt text; tag table headers
- Use accessible colors; compile with LuaLaTeX

Key difference:

- Articles: Keep your existing class! Just add accessibility features
- Slides: Switch to `ltx-talk` (similar syntax, but recreate styling)

Both require LaTeX kernel 2025-11-01 or later (see next sections for details).

See "Getting started" sections at the end for detailed steps. For complete LaTeX Tagging Project documentation, see latest detailed instructions.

3 TeX version requirements (both slides and articles)

- Minimum: TeX Live 2023 or later with all packages updated
- Critical: Must have LaTeX kernel 2025-11-01 or later
  - Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
  - Or use Overleaf Labs (see next section)
- Will NOT work: TeX Live 2022 or earlier

4 You CAN use Overleaf (both slides and articles)

- `ltx-talk` requires a very recent **LaTeX kernel** (2025-11-01 or later)
- This is available through Overleaf's Labs program (not standard Overleaf)

4.1 Using Overleaf (2 steps)

1. **Join Overleaf Labs**:
   - Visit https://www.overleaf.com/labs/participate
   - Opt in and enable "Rolling TeX Live releases"
2. **Configure project**:
   - Set TeX Live version to "Rolling TeXLive (labs)" (bottom of list)
   - Set Compiler to LuaLaTeX

Resources: https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs

5 Common requirements for ALL documents

Whether you're creating slides or articles, these requirements are the same:

5.1 Document Metadata

- Every accessible document must begin with `\DocumentMetadata`
- Goes *before* `\documentclass`

```
\DocumentMetadata{
   pdfstandard=a-2u,     % PDF/A-2u format
   pdfstandard=ua-1,    % Add PDF/UA-1 conformance target
   pdfversion=1.7,     % Explicit PDF version
   lang=en-US,           % Language for screen readers
   tagging=on,           % Enable PDF tagging
   tagging-setup={
      math/alt/use,   % Keep math accessibility enabled (screen readers get tagged math)
      math/mathml/AF=false,% Do not embed MathML as PDF Associated Files
      math/tex/AF=false,   % Do not embed TeX source as Associated Files
      math/mathml/sources= % Clear MathML source attachment list
   }
}
```

This configures accessibility, math tagging, and validator-compatibility behavior in one place.
The `tagging-setup` entries keep math accessible while reducing helper attachments that can trigger warnings in some strict conformance validators.

5.2 Images, Tables, and Colors

- **Images:** All images must include alt text descriptions
  - Allows screen readers to describe visual content
  - Even decorative images need marking
  - See dedicated "Figures" section for examples

- **Tables:** Must specify which rows are headers
  - Helps screen readers navigate table structure
  - Essential for data accessibility
  - See dedicated "Tables" section for examples

- **Accessible colors:** WCAG 2.1 requires 4.5:1 contrast ratio

```
\colorlet{AccessibleRed}{red!80!black}
\colorlet{AccessibleGreen}{green!40!black}
% Standard blue is fine as-is
```

5.3 LuaLaTeX Compilation

- You must compile with LuaLaTeX (not pdfLaTeX or XeLaTeX)

- **Why LuaLaTeX?**
  1. **Automatic MathML:** Makes math accessible without manual work
  2. **Full tagging support:** Complete PDF accessibility features
  3. **Modern fonts:** Handles Unicode and OpenType fonts

- **How to switch:**
  - Command line: `lualatex filename.tex`
  - Most editors: Select "LuaLaTeX" from compiler menu

6 The basics

- Use standard LaTeX environments: `section`, `subsection`, `itemize`, `enumerate`, etc.
- **Existing source files don't need a lot of editing**
- Here's some gratuitous $math$ for the accessibility checker

7 Figures

When including a figure, you must provide alt text describing the image for screen readers. This is *required* for accessibility compliance.

```
\includegraphics[height=.4\textheight,alt={A capybara}]{capybara.jpg}
```

Figure 1: A picture of a capybara

8 Tables

When including a table, you must specify which rows are headers. Screen readers use this information to help users navigate table content. Use `\tagpdfsetup{table/header-rows={...}}` *before* the `tabular` environment:

- Use `{1}` for 1 header row
- Use `{1,2}` for 2 header rows
- Use `{1,2,3}` for 3 header rows, etc.
- **Validator note:** PAC can sometimes report `Table header cell has no associated subcells` even when headers are tagged correctly. The LaTeX team documents this behavior at https://github.com/latex3/tagging

8.1 Example: Table with 3 Header Rows

```
\tagpdfsetup{table/header-rows={1,2,3}}
\begin{tabular}{ccccrcccr}
\toprule
← 3 header rows
\midrule
← data rows
\bottomrule
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

Table 1: A table

9 Common pitfalls

- Confusing PDF tagging with PDF bookmarks
  - `tagging=on` creates structure tree for screen readers (required for accessibility!)
  - PDF bookmarks (navigation outline in left pane) are separate and optional
  - Bookmarks do NOT generate automatically from `tagging=on`
  - For articles: use `\tableofcontents` or the `bookmark` package
  - For slides: See custom bookmark code in `accessible_slides.tex` preamble (Part 2)
- Forgetting alt text for images
  - Every `\includegraphics` needs an `alt={...}` parameter
  - For purely decorative images, use empty alt text: `alt={}`
- Not specifying table header rows
  - Add `\tagpdfsetup{table/header-rows={...}}` before each table
  - Use `{1}` for 1 header row, `{1,2}` for 2 header rows, etc.

- Insufficient color contrast
  - WCAG 2.1 requires 4.5:1 contrast ratio for normal text
  - Avoid light colors: `yellow`, `cyan` fail contrast requirements
  - Darken `red` and `green`: use `red!80!black`, `green!40!black`
  - Standard `blue` is fine and meets WCAG requirements
  - Test with a contrast checker: https://webaim.org/resources/contrastchecker/
- Using the wrong compiler
  - Make sure your editor is set to use LuaLaTeX, not pdfLaTeX
- Old TeX distribution
  - TeX Live 2022 or earlier won't work
  - Update packages using TeX Live Manager (Windows) or TeX Live Utility (Mac)

10 What to ADD to existing documents

When converting existing LaTeX files to be accessible, here's what you need to add:

10.1 Required additions for ALL documents

1. **At the very top** (before `\documentclass`):

```
\DocumentMetadata{
   pdfstandard=a-2u,
   pdfstandard=ua-1,
   pdfversion=1.7,
   lang=en-US,
   tagging=on,
   tagging-setup={...see template...}
}
```

2. **In every** `\includegraphics` command:

```
\includegraphics[alt={Description of image}]{file}
```

3. **Before every table**:

```
\tagpdfsetup{table/header-rows={1}}
\begin{tabular}{...}
```

4. **Change your compiler to LuaLaTeX** (not pdfLaTeX or XeLaTeX)

10.2 Additional for existing Beamer slides

- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- Remove all Beamer themes, color schemes, and template commands
- Keep your `\begin{frame}` environments as-is
- Recreate styling using standard LaTeX packages (xcolor, fontspec, etc.)
- This is one-time preamble work—then reuse for all future talks!

11 Getting started: Common steps for both

1. **Setup environment:** Use Overleaf Labs OR install/update TeX Live locally
   - Overleaf: See earlier section for Labs setup
   - Local: Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
2. **Get templates:** Download from https://github.com/rhstanton/accessible_LaTeX
3. **Add accessibility features:**
   - Add `alt` text to images
   - Add `table/header-rows` to tables
4. **Set compiler to LuaLaTeX**
5. **Compile:**
   - Compile with LuaLaTeX

12 Getting started: What's different by document type

12.1 For Articles (using `article`, `report`, `book`, etc.)

- Add `\DocumentMetadata` before `\documentclass`
- Include the `tagging-setup` block in `\DocumentMetadata` (as shown in this template)
- Switch to `fontspec` and `unicode-math` packages
- Keep your existing documentclass!

12.2 For Slides (migrating from Beamer)

- Copy preamble from `accessible_slides.tex`
- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- Remove Beamer themes/templates/colors
- Recreate styling using standard LaTeX/xcolor
- One-time preamble work—then reuse for all future talks!

13 Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- A practical workflow is to run more than one checker and compare findings.

Checkers used so far: Ally (the accessibility checker built into bCourses), veraPDF (open-source PDF conformance validator, https://verapdf.org/), PAC (PDF Accessibility Checker, https://pac.pdf-accessibility.org/), and Acrobat.

14 Optional validator cleanup step: `strip_af.py`

- The metadata settings alone are enough to generate a 100% score on Ally.
- Stricter checkers (e.g., veraPDF) may still complain about /AF or embedded-file structures.
- Optional strict-archival cleanup command:

  ```
  python strip_af.py build/yourfile.pdf
  ```

- Then validate `build/yourfile.pdf` with `veraPDF`.
- Prerequisite for the script: `python -m pip install pikepdf`

15 Validator-specific notes

- **Not all validator warnings indicate real source problems.**
- **Acrobat: "Lbl and LBody – Failed"**
  - Often appears around figure captions, section numbers, and table captions
  - This is usually an Acrobat checker bug, not a source-tagging error
  - The `<Lbl>` tag is valid for numbering
  - Reference: https://tex.stackexchange.com/a/709655/2388
- **PAC table warning (known behavior)**
  - PAC can report `Table header cell has no associated subcells` even when `table/header-rows` is correct
  - The LaTeX Tagging Project documents this behavior at https://github.com/latex3/tagging-project/issues/1056
- **When this warning appears**
  - First confirm source markup is correct (especially `table/header-rows`)
  - Keep source tagging and document the validator warning
  - Report minimal reproductions to tool vendors when possible
- **Interpretation tip**: Treat one-tool warnings as signals to investigate, not automatic proof of source errors.

Questions or suggestions? richard.stanton@berkeley.edu